Applications of Trigonometry (TE6)

Math 172 lecture notes · Section 4.3.3

Learning objective: solve applications that require the use of trigonometry to find a side or angle.

Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes Text from the handout Extra note (not in your handout)

Overview

FROM THE HANDOUT

Trigonometry is the branch of mathematics that focuses on the relationships between the angles and sides of triangles. It has a wide range of applications across various fields, including science, engineering, architecture, and more. In this section, we will look at some common ways trigonometry is used.

Rounding

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Decimals shown in the middle of a problem are shortened so they're easy to read. Your calculator should keep the ***full value*** *until the final answer. Rounding too early can change the answer.*

Activity 1

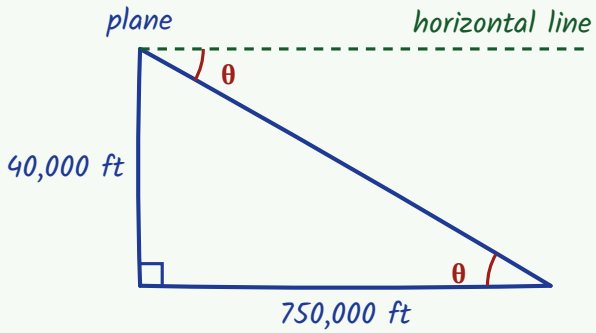
Activity 1 · from your notes

A pilot signals to an air traffic controller that she wants to land. She wants to know what angle of descent to take when she is currently at 40,000 feet. Her plane is a horizontal distance of 750,000 feet from the runway, as the air traffic controller can see on the radar.

(a) *Draw a diagram to represent this situation.*

Answer (a)

A right triangle: the vertical leg is the plane's height (40,000 ft), the horizontal leg is the ground distance (750,000 ft), and the flight path down to the runway is the hypotenuse.



(b) Show the angle of descent in your diagram.

> **Answer (b)**
>
> The angle of descent is at the plane, between a horizontal line through the plane and the downward flight path to the runway. Notice it is **outside** the triangle.

(c) Draw a line that represents the line of sight of the pilot. What do you notice about this line and the base of your triangle?

> **Answer (c)**
>
> The pilot's straight-ahead line of sight is a horizontal line through the plane. (Her line of sight to the runway is the flight path, which slopes down.)
>
> **The horizontal line is parallel to the ground (the base)**

(d) The pilot's line of sight is parallel to the base, so the hypotenuse is a transversal cutting two parallel lines. Which angle of the triangle is congruent to the angle of descent?

> **Answer (d)**
>
> Alternate interior angles are congruent:
>
> **The angle at the runway, between the ground and the flight path**

(e) Find the measure of the angle the pilot needs to make her initial descent.

> **Hint:** Which two sides do you know, relative to that angle?

Answer (e)

From the runway angle, 40,000 is opposite and 750,000 is adjacent, so use tangent:

$$\theta = \tan^{-1}\left(\frac{40{,}000}{750{,}000}\right) \approx 3.05°$$

$$\theta \approx 3.05°$$

A gentle 3° glide makes sense for a landing.

Note

FROM THE HANDOUT

Notice that the angle that we needed to find was an angle that was not inside the right triangle. In these cases, it is helpful to use prior knowledge of parallel lines and angle relationships to determine which other angle is congruent to that given angle.

Definition

FROM THE HANDOUT

Angle of elevation and angle of depression. Many applications of trigonometry include the angle of elevation and the angle of depression, which are formed by two parallel lines cut by a transversal.

- An **angle of elevation** is the angle formed by a horizontal line and a line of sight to a point above the line.
- An **angle of depression** is the angle formed by a horizontal line and a line of sight to a point below the line.

The angle of elevation and the angle of depression are congruent. Because both are formed by parallel horizontal lines, the angle of elevation is congruent to the angle of depression (by the alternate interior angles theorem).

Activity 2

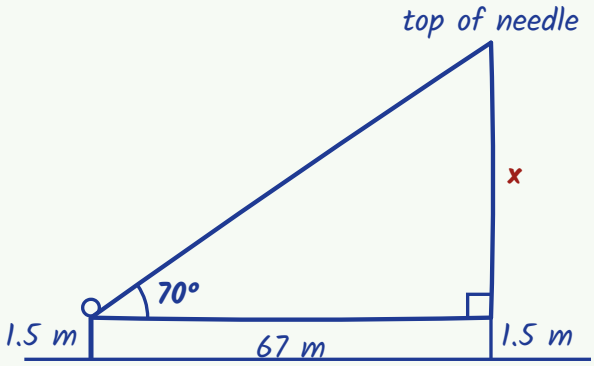
Activity 2 · from your notes

An observer who is 1.5 meters tall is standing 67 meters away from the Seattle Space Needle.

(a) Draw a diagram to represent this situation.

Answer (a)

A right triangle starting at the observer's eyes: 67 m across, and the vertical leg x up to the top of the needle. The extra 1.5 m sits below the triangle.



(b) If the angle of elevation from the observer to the top of the Space Needle is 70°, which trig function could you use to find the height?

Answer (b)

From the 70° angle, x is opposite and 67 is adjacent.

Tangent

(c) How tall is the Space Needle (to the nearest meter)?

Hint: Don't forget to account for the height of the observer!

Answer (c)

$x = 67 \tan 70° \approx 184.08$ m, then add the observer's eye height: $184.08 + 1.5 = 185.58$ m

≈ 186 m

Activity 3 · from your notes

For each problem, draw a diagram and solve.

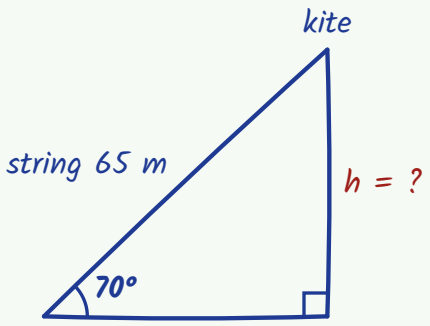
(a) Sarah's kite is flying above a field at the end of 65 meters of string. If the angle of elevation to the kite measures 70°, how high is the kite above Sarah's head?

Answer (a)

The string is the hypotenuse and the height is opposite 70°, so use sine. (We assume Sarah holds the string at about head height, so the height above her hand is the height above her head.)

$h = 65 \sin 70° \approx 61.08$ m

≈ 61 m



(b) Standing on a cliff 380 meters above the sea, Sean sees an approaching ship and measures its angle of depression, obtaining 9°. How far from shore is the ship (to the nearest meter)?

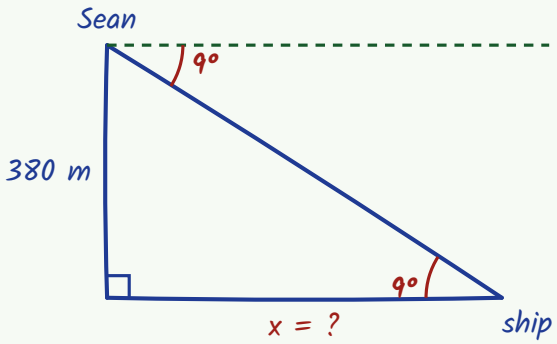
> **Hint:** The 9° is at the top of the cliff, outside the triangle. Which angle inside the triangle matches it?

Answer (b)

The angle of depression (at Sean) equals the angle of elevation at the ship. From the ship, 380 is opposite and x is adjacent:

$$\tan 9° = \frac{380}{x} \rightarrow x = \frac{380}{\tan 9°} \approx 2399.2 \text{ m}$$

≈ 2,399 m



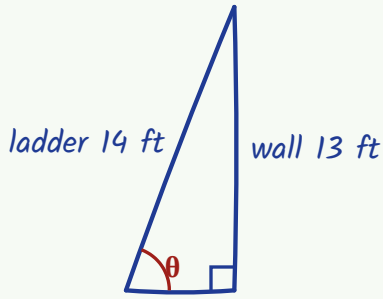
(c) A 14-foot ladder is used to scale a 13-foot wall. At what angle of elevation (to the nearest degree) must the ladder be situated in order to reach the top of the wall?

Answer (c)

Opposite = 13 (the wall) and hypotenuse = 14 (the ladder):

$\theta = \sin^{-1}\left(\frac{13}{14}\right) \approx 68.2°$

$\approx 68°$



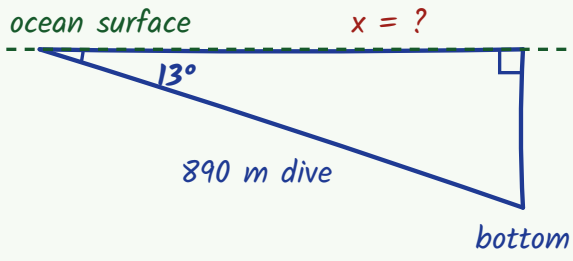
(d) A submarine starts on the surface and dives at an angle of 13° to the surface. It goes diagonally a distance of 890 meters before reaching the bottom. How far is it along the ocean surface from where it started to the point directly above where it reached the bottom?

Answer (d)

890 m is the hypotenuse, and the surface distance is adjacent to 13°, so use cosine:

$x = 890 \cos 13° \approx 867.19 \text{ m}$

≈ 867 m



Note

FROM THE HANDOUT

All applications we have done so far have involved right triangles. Let's now look at some examples of non-right triangle applications.

Activity 4

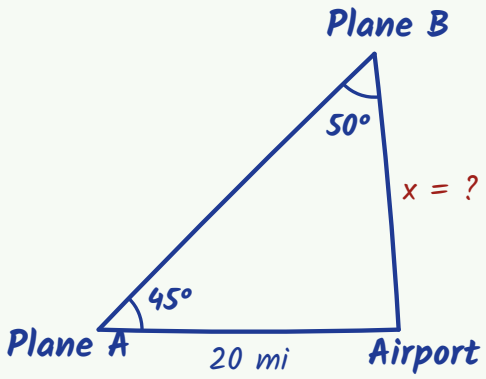
Activity 4 · from your notes

Airplane A is flying directly towards the airport, which is 20 miles away. From Airplane A's point of view, the angle between the airport and Airplane B is 45°, and from Airplane B's point of view, the angle between the airport and Airplane A is 50°.

(a) Draw a diagram to represent this situation.

Answer (a)

A non-right triangle with corners Plane A, Plane B and the Airport. The 20 miles runs from Plane A to the Airport.



(b) How far is Airplane B from the airport (to the nearest tenth of a mile)?

> **Hint:** Which angle is across from the side you want? Which angle is across from the 20 miles?

Answer (b)

The side we want (B to the airport) is across from the 45° angle. The 20 mi side is across from the 50° angle. That's two complete pairs, so use the Law of Sines:

$$\frac{x}{\sin 45°} = \frac{20}{\sin 50°} \rightarrow x = \frac{20 \sin 45°}{\sin 50°} \approx 18.46 \text{ mi}$$

≈ 18.5 miles

Activity 5

Activity 5 · from your notes

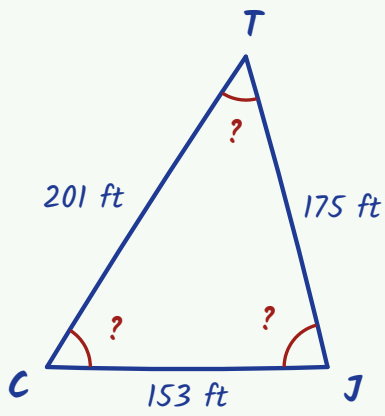
Carlos, Jean, and Travis are camping in their tents. The distance between Carlos and Jean is 153 feet, the distance between Carlos and Travis is 201 feet, and the distance between Jean and Travis is 175 feet.

(a) Draw a diagram to represent this situation.

> **Hint:** It might be helpful to label your triangle with C, J, and T for the angles and c, j, and t for the sides.

Answer (a)

Label angles C, J, T at each person's tent. Each side gets the lowercase letter of the angle across from it: $c = 175$ (Jean–Travis), $j = 201$ (Carlos–Travis), $t = 153$ (Carlos–Jean).



(b) What type of triangle represents this situation? How can you determine whether this triangle is a right triangle?

Answer (b)

Test the Pythagorean Theorem with the longest side: $153^2 + 175^2 = 54{,}034$, but $201^2 = 40{,}401$. They don't match.

Not a right triangle: an SSS (side-side-side) triangle

(c) Which law (Sines or Cosines) would be best to find the angle at Carlos?

Answer (c)

We know all three sides but no angles, so there's no complete pair.

Law of Cosines

(d) Find the angle at which Carlos is located from his friends, to the nearest degree.

Answer (d)

$$\cos C = \frac{t^2 + j^2 - c^2}{2tj} = \frac{153^2 + 201^2 - 175^2}{2(153)(201)} = \frac{33{,}185}{61{,}506} \approx 0.5395$$

$C \approx 57.35°$

$C \approx 57°$

(e) Now find the angle at Travis (to the nearest degree) using the Law of Sines.

Answer (e)

$$\frac{\sin T}{153} = \frac{\sin 57.35°}{175} \rightarrow \sin T \approx 0.7361 \rightarrow T \approx 47.40°$$

$T \approx 47°$

(f) Find the remaining angle of the triangle.

Answer (f)

$J = 180° - 57.35° - 47.40° \approx 75.25°$

$J \approx 75°$

⚠ **Watch out:** Subtracting the rounded angles (180 − 57 − 47) gives 76°, which is off by one. The textbook's key lists 76° for this reason. Subtract unrounded values, then round.

Note

FROM THE HANDOUT

Trigonometric functions can model relationships between quantities that follow a periodic pattern: height over time, distance over time, temperature over time, and so on. Scientists observe this back-and-forth movement and collect data so they can model it using an equation or a graph. They then use this information to make predictions for the future.

Activity 6

Activity 6 · from your notes

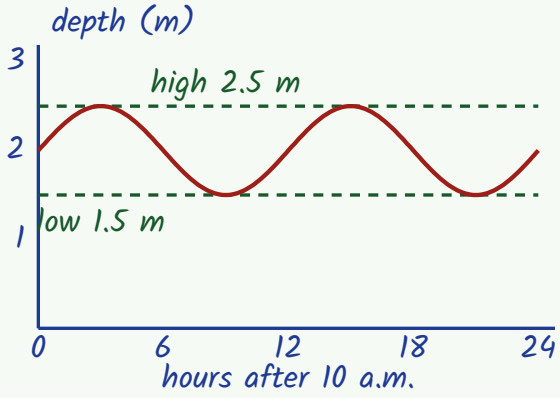
The depth of the water in meters at a certain pier varies with the tide and is modeled by $d(t) = 2 + \frac{1}{2}\sin(\frac{\pi}{6}t)$, where t is the number of hours after 10 a.m.

(a) How deep will the water be at high tide?

Answer (a)

Sine swings between -1 and 1, so $\frac{1}{2}\sin(\ldots)$ swings between $-\frac{1}{2}$ and $\frac{1}{2}$. The highest depth is $2 + \frac{1}{2}$.

2.5 m



(b) How deep will it be at low tide?

Answer (b)

The lowest depth is $2 - \frac{1}{2}$.

1.5 m

(c) How many hours are there between two successive high tides?

Hint: The time from one high tide to the next is one full cycle. What's the period of this sine function?

Answer (c)

That's the period: $\frac{2\pi}{b} = \frac{2\pi}{\pi/6} = 2\pi \cdot \frac{6}{\pi} = 12$

12 hours

(d) How many of these tide cycles are there in a 24-hour day?

Answer (d)

$\frac{24}{12} = 2$

2 cycles

Activity 7 · from your notes

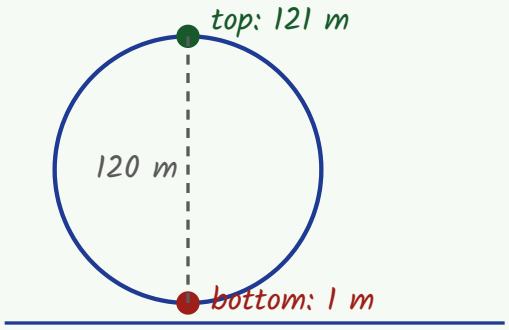
A circular Ferris wheel is 120 meters in diameter and contains several carriages. Jesus and Allison enter a carriage at the bottom of the wheel and get off 24 minutes later after having gone around 8 times. When a carriage is at the bottom of the wheel, it is 1 meter off the ground.

(a) What are the maximum and minimum heights of Jesus and Allison's carriage?

Answer (a)

The bottom is 1 m off the ground. The top is one full diameter higher: $1 + 120$.

Maximum 121 m, minimum 1 m



(b) Let $h(t)$ be the height of the carriage t minutes after it starts moving. What is the period of $h(t)$?

Answer (b)

8 trips around take 24 minutes, so one trip takes $\frac{24}{8}$.

3 minutes

(c) Which trigonometric function would be the best to use to model this situation?

> **Hint:** Where does the ride start: at the top, the middle, or the bottom? Which graph starts there?

Answer (c)

The ride starts at the minimum height. Cosine starts at its maximum, so a cosine flipped upside down (negative cosine) starts at its minimum.

A reflected (negative) cosine function

For example, $h(t) = -60\cos\left(\frac{2\pi}{3}t\right) + 61$, which gives $h(0) = 1$ and a high of 121.

Practice Example 8

Practice Example 8 · from your notes

From a point on the ground 50 feet from the base of a building, the angle of elevation to the top of the building is 38°. Find the height of the building, rounded to two decimal places.

Try it, then check.

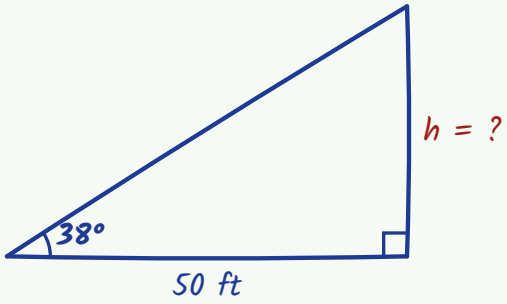
Answer

Height is opposite 38° and 50 ft is adjacent:

$h = 50\tan 38° \approx 39.064\ldots$

$h \approx 39.06$ ft

The textbook shows 39.07 because it rounds tan 38° to 0.7813 first. Keep full calculator precision until the end.



Practice Example 9

Practice Example 9 · from your notes

A drone hovers at a height of 120 feet. The angle of depression from the drone to a landmark on the ground is 25°. Find the horizontal distance from the point directly below the drone to the landmark, rounded to two decimal places.

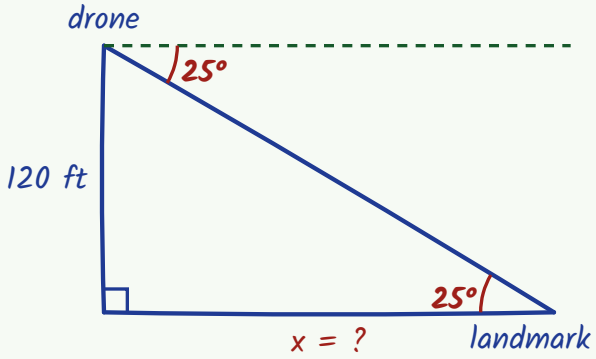
Try it, then check.

Answer

The angle of depression equals the angle of elevation at the landmark. From there, 120 is opposite and x is adjacent:

$$x = \frac{120}{\tan 25^\circ} \approx 257.340\ldots$$

$x \approx 257.34$ ft



Practice Example 10

Practice Example 10 · from your notes

Two people stand on opposite sides of a flagpole. One measures the angle of elevation to the top as 42° from a distance of 30 feet; the other stands 45 feet away on the opposite side. Find the second person's angle of elevation, rounded to one decimal place, using the flagpole's height found from the first measurement.

Try it, then check.

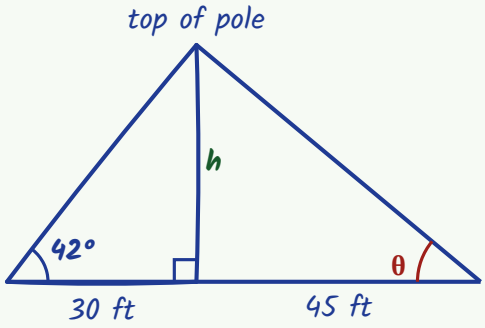
Answer

Height: $h = 30 \tan 42° \approx 27.012$ ft

Second angle: $\theta = \tan^{-1}\left(\frac{27.012}{45}\right) \approx 30.98°$

$\theta \approx 31.0°$

Check: the farther person sees a smaller angle. $31.0° < 42°$ ✓



Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture.

Accessibility & help

These notes are built to work with screen readers, keyboard-only use, zoom and phone screens. Every answer stays hidden until you open it, and **Switch to standard font** at the top changes the handwriting to a plain font.

Other formats: If you need these notes another way (large print, a Word file, braille, or math written in MathML), message your instructor through Canvas Inbox, or contact Santa Ana College Disabled Students Programs & Services (DSPS): sac.edu/DSPS, DSPS@sac.edu, (714) 564-6295.

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